Natural Language Processing

Autumn 2026

Week 4: Named Entity Recognition

NLP: A profusion of unstructured data

Natural language processing (NLP) is at the intersection of linguistics and machine learning, concerned with the **understanding** and generation of unstructured natural languages.

Common unstructured text data sources:

- Documents within an organisation
- Web pages
- Social media
- Publications, news media transcripts
- Speech-to-text (Siri, Cortana, Alexa…)

Named Entities

Named Entity Types:

- Person (e.g., Elon Musk)
- Organization (e.g., OpenAI)
- Location (e.g., New York)
- Date/Time (e.g., March 5, 2025)
- Miscellaneous (e.g., Tesla Model 3, Bitcoin)

Challenges in NER

- Ambiguity (e.g., "Amazon" as a river vs. a company)
- Context dependency
- Handling unseen entities

Named Entity Recognition

- **Named Entities (NE)** are words or phrases that refer to specific real-world entities such as persons, organizations, locations, dates, times, and numerical quantities.
- NE Recognition (NER) is a sub-task of Information Extraction (IE)
- NER is to process a text and identify named entities
  - e.g. “U.N. official Ekeus heads for Baghdad.”

[ORG U.N.] official [PER Ekeus] heads for [LOC Baghdad] .

- NER is also an important task for texts in **specific domains** such as biomedical texts

Difficulties with NER

- Names are too numerous to include in dictionaries
- Variations
  - e.g. John Smith, Mr Smith, John
- Ambiguities
  - Same name refers to different entities, e.g.
    - JFK – a former US president
    - JFK – his son
    - JFK – airport in NY
- Limited labelled training data for some languages
- Domain Adaptation Issues
  - NER models trained on general data may not work well in specific domains
  - Example: Medical vs. legal vs. social media data
  - Need for domain-specific annotated corpora

Named-Entity Recognition (NER)

- **NER** aims to detect all the **pre-defined named entities** in a given text.

Typical named entities:

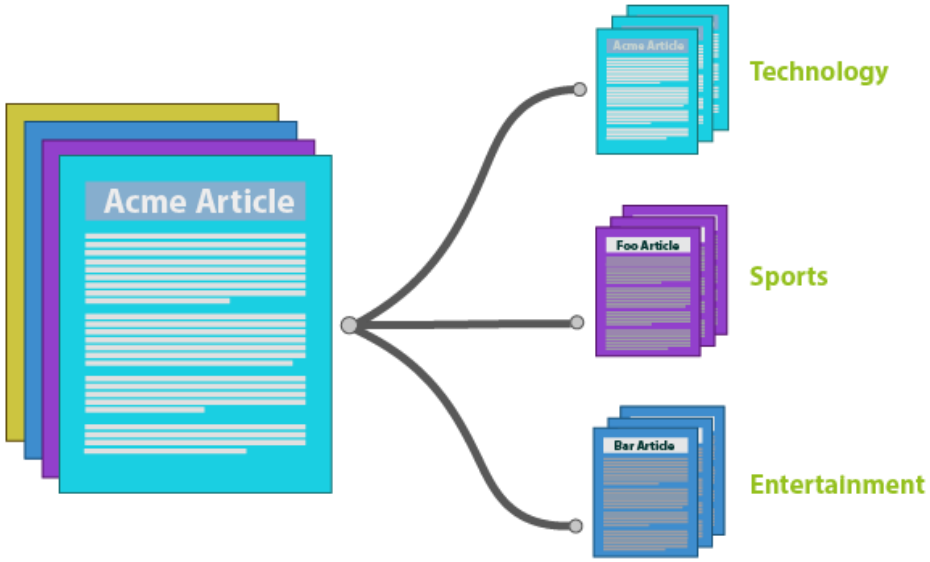
- Person names
- Company/Organizations
- Locations (street, cities, states, countries…)
- Dates
- Money
- Titles (Mr/Ms, Dr/Dra, CEO, president…)
- Products/Brands
- Numbers
- URLs
- Phone numbers

And many more!

NER Applications

NER is **a pillar task** of many higher-level NLP applications

- **<u>Document classification/clustering</u>**:



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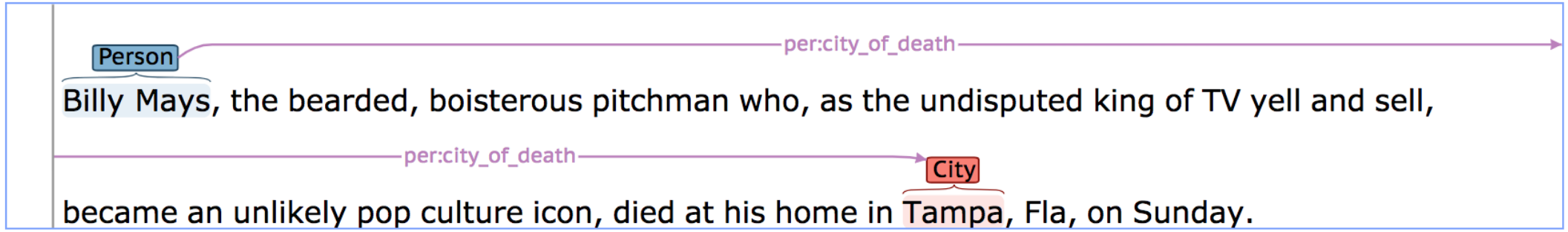
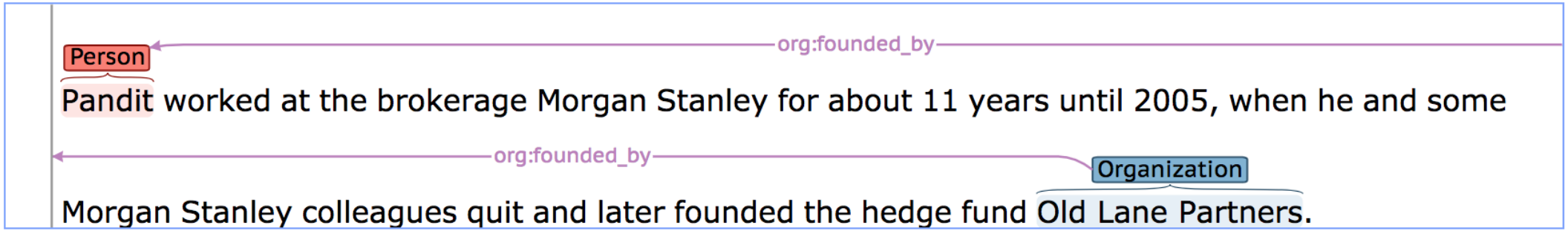
NER Applications

NER is **a pillar task** of many higher-level NLP applications

- **<u>Entity relation extraction</u>**: For example: Where did Billy Mays pass away?



What is the name of the hedge fund that Vikram Pandit co-founded?



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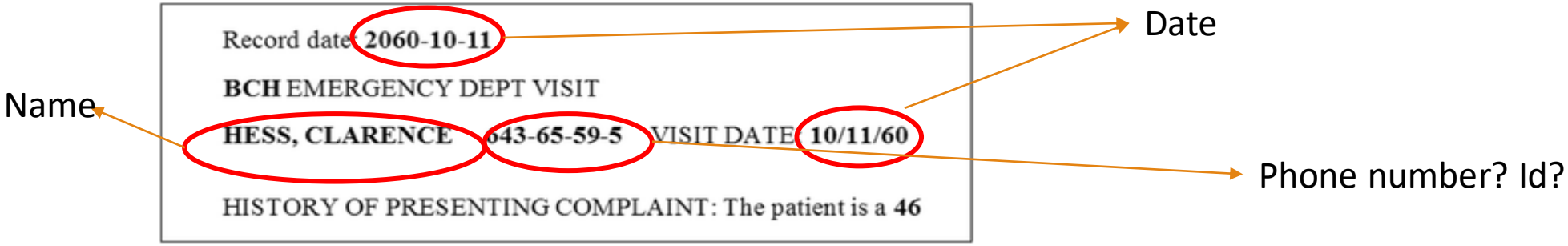
NER Applications

NER is **a pillar task** of many higher-level NLP applications

- **<u>Document de-identification</u>**:
  - Mask personally identifiable information (PII)
  - E.g. Research with **electronic health records**. It is **vital** to remove any personal information/data to preserve patients' privacy



NER Applications

NER is **a pillar task** of many higher-level NLP applications

- **<u>Task-oriented dialogue systems</u>**:

<u>Dialogue system</u>

Slot filling:

Date: 15th of June
Departure city: Beijing
Arrival city: Berlin

Form query (e.g. SQL):

SELECT * FROM *table* WHERE…

I want to book a flight on the **15th of June** from **Beijing** to **Berlin**.

Reply with results

These are the available flights from Beijing to Berlin the 15th of June:
…….

NER Tagging

INPUT →	*My*	*name*	*is*	*Michael*	*Jordan*	*and*	*I*	*live*	*in*	*Sydney*	*.*
OUPUT ←	*O*	*O*	*O*	*B-PER*	*I-PER*	*O*	*O*	*O*	*O*	*B-LOC*	*O*

Sequential classification task

Multi-token named entities (e.g. Michael Jordan , Sydney …)

- BIO tagging (B-> Beginning, I-> Inside, O-> Outside)

Beyond BIO – Advanced Tagging Schemes

- The standard BIO (Begin, Inside, Outside) scheme is effective but can be blunt. It uses the same tag (**I**) for the middle, the end, and even the second word of a two-word entity. This lack of "boundary awareness" makes it harder for a model to learn exactly where an entity should stop.
- Solution: The BILOU / BIOES Scheme To provide the model with more structural "hints," we expand the tag set:
  - B (Begin): First token of a multi-token entity.
  - I (Inside): Middle tokens of a multi-token entity.
  - L / E (Last/End): The final token of a multi-token entity.
  - O (Outside): Non-entity tokens.
  - U / S (Unit/Single): An entity consisting of exactly one token.

Function	**BILOU**	**BIOES**
Beginning	**B** (Begin)	**B** (Begin)
Inside	**I** (Inside)	**I** (Inside)
Last / **E**nd	**L** (Last)	**E** (End)
Outside	**O** (Outside)	**O** (Outside)
Unit / **S**ingle	**U** (Unit)	**S** (Single)

Compare BIO with BILOU

- Consider this example: "Elon Musk visited Tesla in Texas."

Token	BIO (Standard)	BILOU (Advanced)	Why BILOU is more precise
Elon	B-PER	**B-PER**	Both mark the **Beginning**.
Musk	I-PER	**L-PER**	BIO uses "Inside," but BILOU uses **Last**. This tells the model "this is the end of the name."
visited	O	O	Both mark as Outside.
Tesla	B-ORG	**U-ORG**	BIO uses "Begin," but BILOU uses **Unit**. This tells the model "this is a complete one-word entity."
in	O	O	Both mark as Outside.
Texas	B-LOC	**U-LOC**	Again, BILOU identifies this as a standalone **Unit**.

The Real-world is not flat: Nested Entities

Standard NER assumes a “flat” structure (one tag per token). However, language is often hierarchical and nested.

For example: “University of Technology Sydney" contains a Location (Sydney) inside an Organization.

Solutions:

- Layered/Stacked Classifiers: Pass the output of one NER layer into another to detect outer entities.
- Biaffine Parsers: Treat NER as a dependency parsing problem, predicting the probability of a span between any two tokens being an entity.

Layered / Stacking Approach

In this architecture, you run multiple passes of an NER model. The first layer identifies the innermost (smallest) entities, and the second layer uses those results as features to find the outer entities.

Example: "The World Health Organization headquarters in Geneva"

- **Layer 1 (The Smallest Unit):** Identifies "Geneva" as GPE (Geopolitical Entity).
- **Layer 2 (The Containing Unit):** Identifies "World Health Organization" as ORG.
- **Layer 3 (Global):** Can identify the entire span "World Health Organization headquarters in Geneva" as a FACILITY.

Biaffine Parsing (Span-based Prediction)

Instead of tagging individual tokens (BIO), this method looks at every possible **span** (start and end point) in a sentence and calculates a score for whether that span is an entity.

Example: "The [[North Atlantic] Treaty Organization]"

- The model evaluates the span **[North Atlantic]**: It assigns a high score for LOC.
- The model evaluates the span **[North Atlantic Treaty Organization]**: It assigns a high score for ORG.

Result: Because the model scores *spans* rather than *tokens*, it can easily output **two entities** for the same set of words.

Public Benchmarks for NER

- CoNLL-2002 and CoNLL-2003 (British newswire)
  - Multiple languages: Spanish, Dutch, English, German
  - 4 entities: Person, Location, Organization, Misc
- MUC-6 and MUC-7 (American newswire)
  - 7 entities: Person, Location, Organization, Time, Date, Percent, Money
- ACE
  - 5 entities: Location, Organization, Person, FAC, GPE
- BBN (Penn Treebank)
  - 22 entities: Animal, Cardinal, Date, Disease, …

CoNLL-2003

CoNLL – Conference on Natural Language Learning by the ACL's Special Interest Group on Natural Language Learning

Shared Task: Language-Independent Named Entity Recognition

Goal: Identify boundaries and types of named entities

- 4 entity types: Person, Organization, Location, Misc.

Data:

- Use the **BIO notation**
  - B- -- beginning of a chunk/NE
  - I- -- internal of a chunk/NE
  - O – outside of any chunk/NE
- 4 pieces of info for each term (Token, POS Tag, BIO Tag, Entity Type)

https://www.nltk.org/book/ch07.html

How do we evaluate the performance of a NER model?

Have you heard “F measure”?

Confusion Matrix

		Predicted Class: Positive	Predicted Class: Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

Image: https://shorturl.at/9iYwE

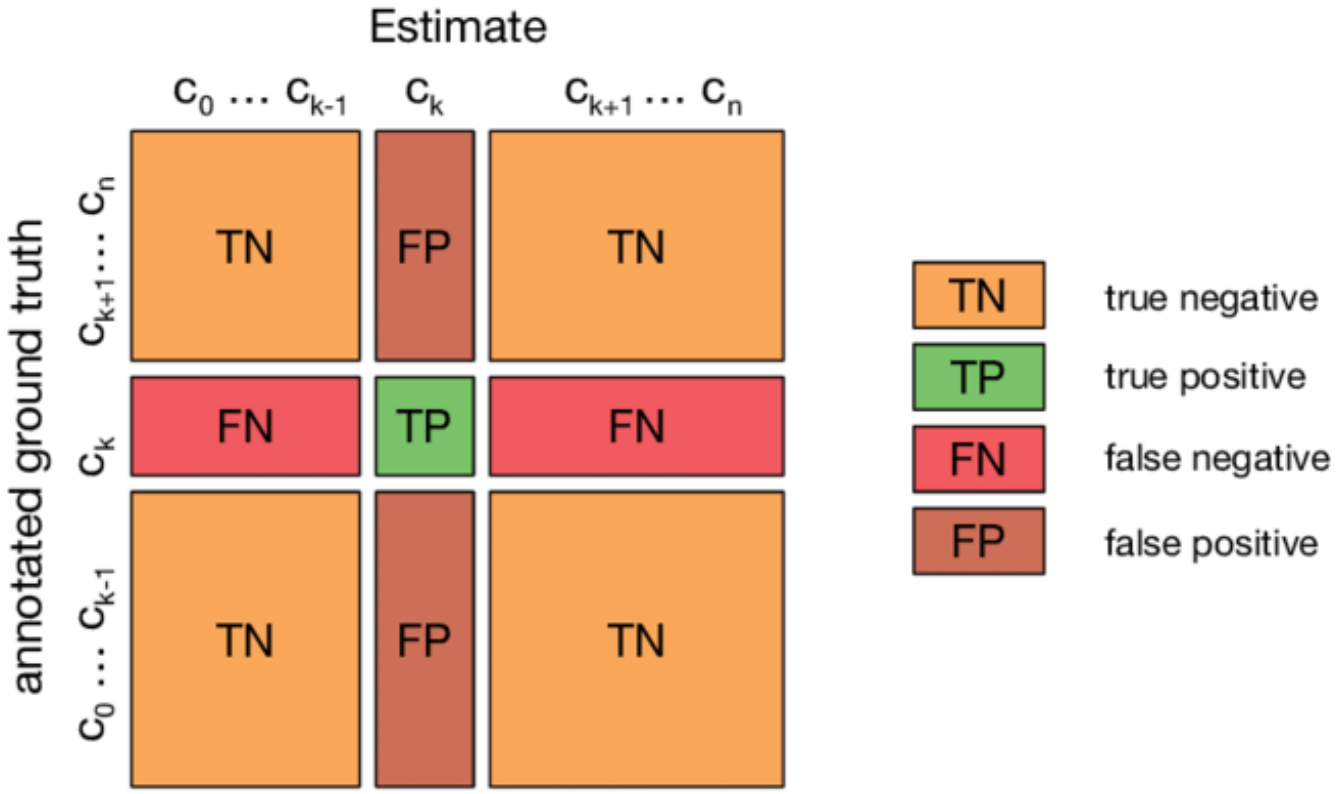
Sensitivity is also called “Recall”.

Confusion Matrix - Example

	Predicted class POSITIVE (spam ✉)	Predicted class NEGATIVE (normal 🕵)	
Actual class POSITIVE (spam ✉)	TRUE POSITIVE (TP) ✉ ✉ **320**	FALSE NEGATIVE (FN) ✉ 🕵 **43**	*Recall* $= \frac{TP}{TP + FN} = \frac{320}{320 + 43} = 0.882$
Actual class NEGATIVE (normal ✉)	FALSE POSITIVE (FP) ✉ ✉ **20**	TRUE NEGATIVE (TN) ✉ 🕵 **538**	
	Precision $= \frac{TP}{TP + FP} = \frac{320}{320 + 20} = 0.941$		

Confusion Matrix for Multiple Classes

When there are multiple classes, usually only one of them is considered as the “Positive” class.



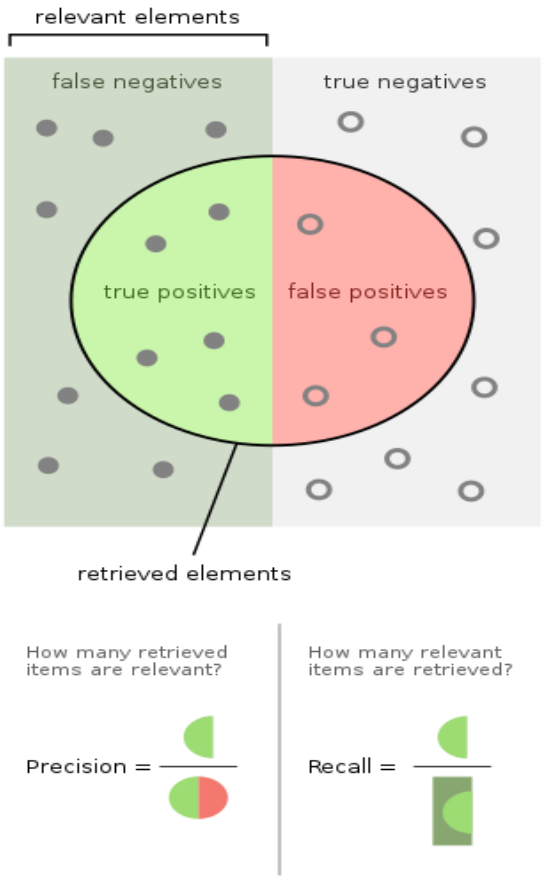
F measure

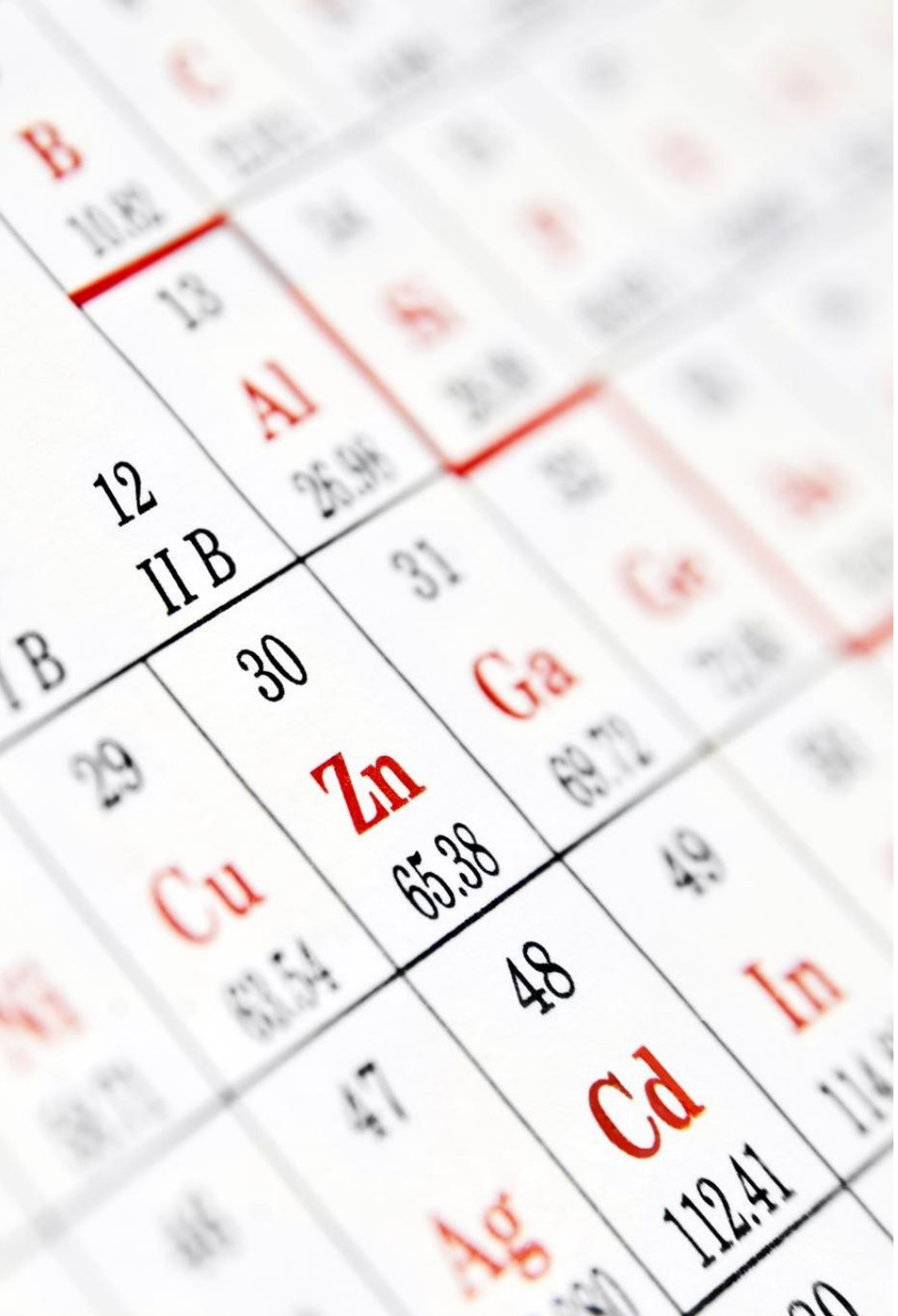
- **F measure** (also called F1 score): Most common metric. Harmonic-mean between **precision** and **recall**

$$precision = \frac{TP}{TP + FP} \qquad recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{precision \times recall}{precision + recall}$$

TP= True Positive, FP = False Positive and FN = False Negative



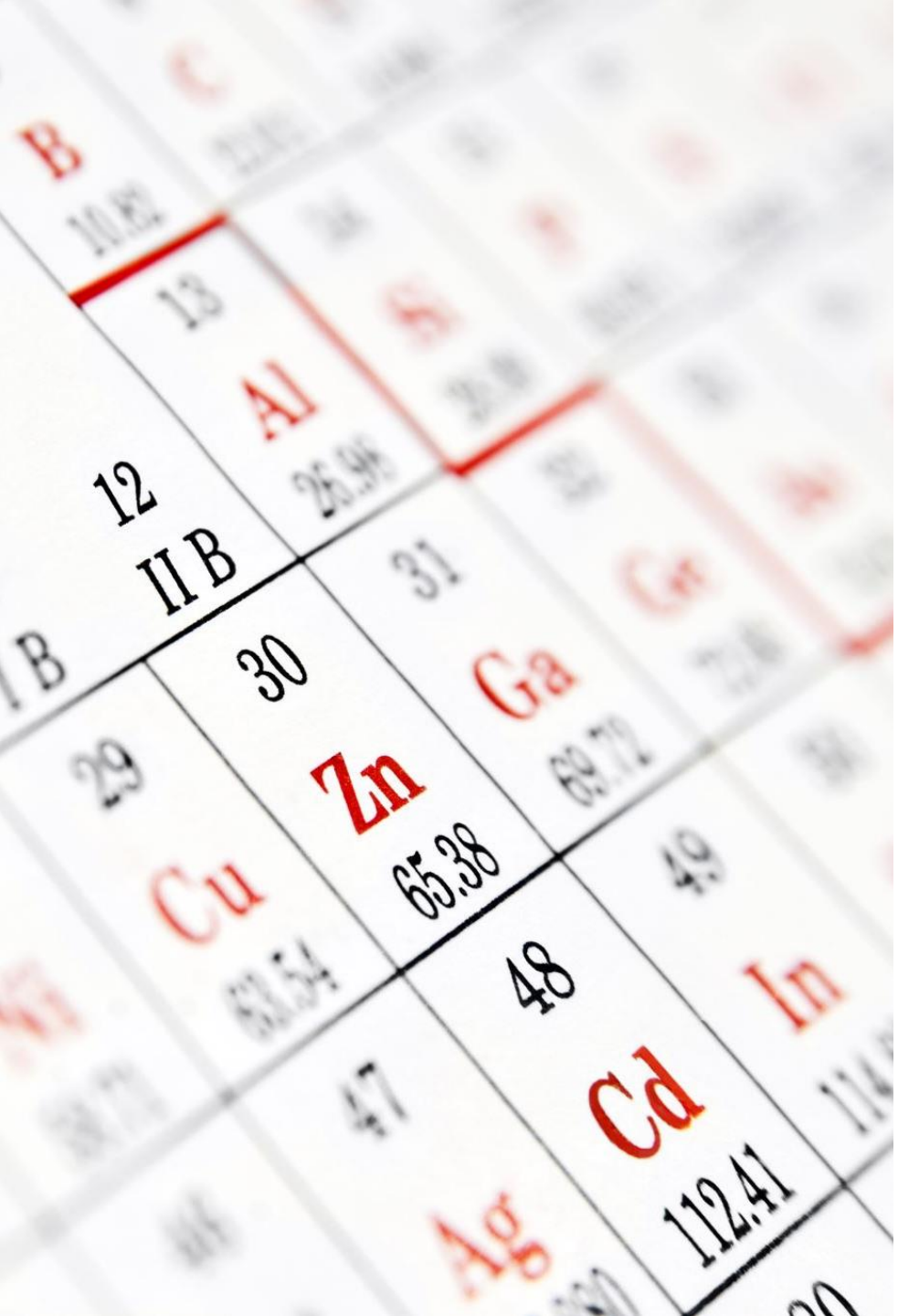


F1 Measure

Scenario 1: Precision = 40% Recall = 40%

Scenario 2: Precision = 75% Recall = 25%

Which one is better?



F1 Measure

Scenario 1: Precision = 40% Recall = 40% F1 = 40%

Scenario 2: Precision = 75% Recall = 25% F1 = 37.5%

Which one is better?

Training NER - Traditional machine learning

Two step approach:

- Feature engineering:
  - **Word casing, word length**, **morphological patterns, ontologies**…
- Train a classifier:
  - **Hidden Markov Model** (HMM) (Bikel et al., 1997), **Support Vector Machine** (SVM) (Masayuki and Matsumoto, 2003), **Conditional Random Field** (CRF) (McCallum and Li, 2003)

Advantages:

- Better generalization:
  - Learning from the data

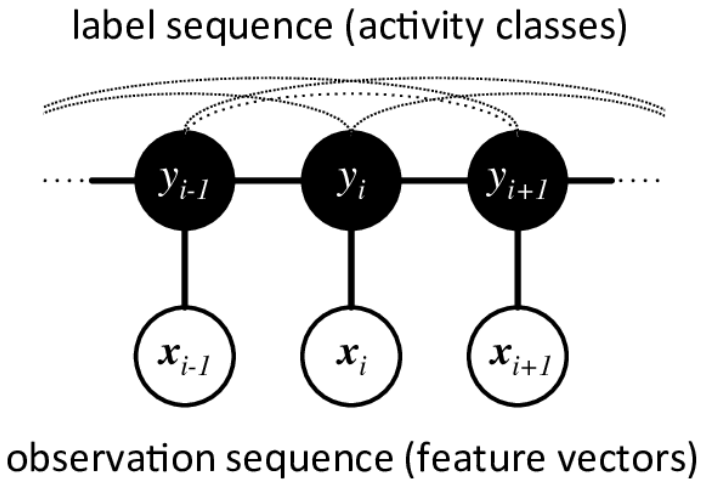
Limitations:

- Features often domain-specific and they are costly to develop them (requires human knowledge)

Conditional Random Fields (CRF)

- In CRF, the structure is often a linear chain, where the label state sequence connects y_{i-1} and y_i, and each state y_i is dependent on the observation sequence x.

- CRFs previously used to be widely used and applied in NLP research, including
  - Part-of-speech tagging
  - Named Entity Recognition
  - Chinese word segmentation
  - etc.



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Training NER - Deep Learning

Currently, most NER models use some kind of deep learning algorithm:

- Recurrent neural networks (RNNs)
  - Long short-term memory networks (LSTMs) (Hochreiter and Schmidhuber, 1997)
  - Gated Recurrent Units (GRUs) (Chung et al., 2014)
- Transformers (Vaswani et al, 2017)

Advantages:

- Learn directly from data (avoid feature engineering)
  - Only features **word/character embeddings**
- Higher accuracy

Limitations:

- They need **more annotated training data**
  - i.e. overfitting problem

State-of-the-art NER models

Deep learning

- Pre-trained language models, such as Transformer-Based NER Models
- **Fine-tune the output layer** of the neural network over a dedicated dataset

Advantages:

- Less annotated data required
- Higher accuracy
- Pre-trained language models widely available (Download from Huggingface.co)

Limitations:

- GPUs required
- Slower inference time

NER Processing Pipeline

Input Text
↓
Tokenization
↓
Context Representation (embeddings)
↓
NER Model
↓
Entity Classification
↓
Structured Output

An Example of a “Structured Output”:

```
{
 "PERSON": ["Elon Musk"],
 "ORG": ["Tesla"],
 "LOCATION": ["Texas"]
}
```

Pre-trained NER Models

Popular pre-trained models:

- spaCy: Industrial-strength NER with pre-trained models.
- Hugging Face: Fine-tuned BERT, RoBERTa, etc.

Applications of NER

Information Retrieval

- Enhancing search engines by recognizing named entities in queries.
- Example: Searching "Who is the CEO of Tesla?" returns Elon Musk as an entity.

Automatic Content Classification

- Classifying articles, blogs, or news reports with relevant named entities.
- Example: News article about "Apple Inc." gets tagged under "Technology, Business."

Applications of NER (cont’)

Healthcare and Biomedical Text Mining

- Identifying diseases, symptoms, and drugs in medical records or research papers.
- Example: Extracting mentions of "COVID-19," "Remdesivir," and "fever" from clinical data.

E-Commerce and Customer Support

- Enhancing recommendation systems by recognizing brands and products in reviews.
- Example: "I love my new Nike Air Jordans!" → Recognizes "Nike" (Brand) and "Air Jordans" (Product).

Questions?